AC 90 Operations and Maintenance Manual

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Fictional equipment and specifications created solely for this hackathon. These documents are a closed demonstration dataset and must not be used to operate or maintain real equipment.

Equipment identity and purpose

The fictional AC-90 is an enclosed air compressor supplying compressed air to a demonstration production line. It is a different machine from the HP-200 hydraulic press. It uses AC-L32 compressor lubricant in this dataset. Do not transfer hydraulic-press values, oil types, or alarm definitions to this compressor.

Operating reference values

Parameter	AC-90 reference
Normal outlet pressure	7.0 to 8.0 bar under steady demand
Pressure-low warning	Below 6.5 bar for more than 10 seconds under steady demand
Discharge temperature normal range	70 to 90 degrees C
High discharge-temperature warning	Above 95 degrees C
High discharge-temperature stop	105 degrees C or higher
Lubricant level indication	Between LOW and HIGH marks

Operating states

READY means the required indications are acceptable before the demonstration run. LOADED means the compressor is supplying air under demand; the normal outlet-pressure range applies to steady demand in this state. HOLD means operation has been interrupted for review. A transient pressure dip is not automatically the defined pressure-low warning.

Document map

Page 2 defines alarm codes and their documented responses. Page 3 lists maintenance intervals and escalation requirements. This single AC-90 manual is the complete reference for M-AC90 in the student pack. No AC-90 supplementary manual is provided.

Alarm Reference and Diagnostic Limits

Code	AC-90 meaning	Documented response
E17	High discharge temperature	Record temperature; above 95 degrees C, hold and request cooling-path review. At 105 degrees C or higher, the high-temperature stop applies.
E08	Low outlet pressure	Below 6.5 bar for more than 10 seconds under steady demand: record demand and pressure; request leak and inlet-filter review.
E31	Condensate-drain fault	Record the drain fault indication and request authorized drain inspection. This is not a guard-open code on AC-90.
E42	Inlet-filter restriction	Record the inlet-filter indicator and review the AF-10 service history.

High discharge temperature or E17

Review the reported cooling-air path condition, cooling-fan status indication, lubricant-level indication, and recent demand pattern. A reading of 98 degrees C exceeds the warning level but is below the stop threshold. Keep the event on HOLD for review. The dataset does not establish the cause from temperature alone.

Low outlet pressure or E08

Record the pressure, duration of the low reading, and demand state. Report visible or audible leak observations without attempting repair, then review the AF-10 restriction indication. A low-pressure event does not by itself prove a worn compressor element. Internal diagnostic procedures are not provided.

Machine-specific codes

E17 on AC-90 means high discharge temperature; E17 on another machine may mean something else. Interpret the code using the selected machine and its own manual. This page does not authorize adjustment of pressure or temperature limits.

Maintenance Schedule and Escalation

When	AC-90 item	Required record
Before each shift	Lubricant level and condensate-drain status	LOW-to-HIGH level indication and drain status
Every 100 operating hours	AF-10 inlet-filter restriction indicator	Indicator state and abnormal findings
Every 400 operating hours	External cooling-air path	Obstruction observation and reviewer
Every 600 operating hours	Planned AF-10 replacement	Part identifier, hours, date, reviewer
Every 3,000 hours or 12 months	Planned AC-L32 lubricant replacement	Whichever limit occurs first; record date and hours

Inspection and replacement are different

The AF-10 indicator is checked every 100 operating hours; planned replacement is every 600 hours. Confirmed restriction can require earlier authorized replacement. A calendar interval for AF-10 replacement is not supplied. The HP-200 RF-20 filter is not the AC-90 filter.

Escalation and review

Record machine ID M-AC90, timestamp, operating hours, active alarm, outlet pressure, discharge temperature, demand state, lubricant indication, filter indication, drain status where relevant, and previous incident reference. Repeated warnings or unresolved abnormal indications remain on HOLD for authorized review.

Physical maintenance requires authorized personnel and the site’s isolation and pressure-release procedure, which is not supplied here. The supervisor records any return-to-service decision. The assistant may summarize documented information but cannot authorize operation.

Information not provided

This manual does not supply maintenance costs, spare-part prices, supplier contacts, electrical diagrams, tightening torques, motor rewinding steps, compressor-element replacement steps, expected remaining life, or a guaranteed repair time. Do not invent these details.